

The background of the slide features a photograph of several wind turbines in a field, with solar panels visible in the foreground. The sky is a mix of orange, yellow, and blue, suggesting a sunset or sunrise. A large, dark blue diagonal shape is overlaid on the right side of the image, creating a modern, geometric design.

Carbon Abatement, Net-Zero, and Sustainable Solutions in Partnership with Industry and Government

Energy and Resources Institute

Introduction

The **Energy and Resources Institute** (ERI) is committed to delivering engineering and technological innovations in the energy and resources sectors to enable industry to develop and adopt carbon-abatement net-zero and sustainable solutions. ERI also carries out R&D to support industries transition their energy requirements to renewable and sustainable energy sources and develop sustainable processes and solutions to reduce their environmental and carbon footprint. We ensure that our research is impactful through our partnerships with Industry and Government.

ERI's research is geared towards leveraging lab-scale innovations and solutions to their commercial realisation through feasibility studies, proof-of-concept studies, pilot-scale demonstration projects, and technical support for commercialisation. Our research expertise and state-of-the-art facilities provide contract research and consultancy services, including analytical and testing services.



Research and Consultancy Services

We provide research and consultancy services in the following areas:

- ➔ **Renewable Energy**
 - > Renewable and Hybrid power systems analysis and modelling.
 - > Hydrogen integration into power systems
 - > Monitoring and predictive maintenance of solar PV systems
 - > Photovoltaics
- ➔ **Waste-to-energy and recycling**
 - > Biomass/plastic wastes to energy and value-added products
 - Pyrolysis
 - Anaerobic digestion
- ➔ **Advanced Manufacturing and Materials Engineering**
 - > Materials Development
 - > Development of industry procedures and standards.
 - > Corrosion engineering in the energy and resources sector
- ➔ **Measurements, Modelling, and Analytics**
 - > IoT systems and cybersecurity
 - > Heating, ventilating, and air conditioning (HVAC) systems
 - > Digitisation of energy and resources sectors – data analytics and artificial intelligence
- ➔ **Process Safety**
 - > Safety evaluations and procedures for chemical processes
 - > Fire and explosion safety
- ➔ **Materials Characterisation and Analytical Services**

Our laboratories are equipped to carry out a variety of physical and chemical analysis of materials.

 - > Analysis of volatile and soluble organic compounds
 - > Organic elemental analysis
 - > Thermogravimetric analysis
 - > Trace analysis
 - > Thermogravimetric analysis
 - > Gas and liquid composition (Chromatography)
 - > Materials strength and hardness testing
 - > Corrosion testing
 - > Particle size and zeta potential
 - > Calorimetry – energy content of materials
 - > Fiber analysis of plant and food materials
 - > PFAS and foams testing





Renewable Energy Grid Testing Facility (REMHART)

REMHART is an industry-focused innovation hub for renewable energy systems. The Hub will focus on enabling affordable, reliable, and green power systems through research collaborations with industry to develop innovative approaches to address challenges in the design, deployment and operation of renewable energy systems. Central to this Hub is the Microgrid facility, located at East Arm in Darwin, underpinning the industry-university collaborations.

We are particularly interested in studies aimed at developing projects of commercial potential in partnership with the industry. This hub will also produce training packages to build a workforce ready to take advantage of the proliferation of renewable energy systems in the Northern Territory (NT).

The Grid Testing Facility is a unique setup that is essentially a surrogate microgrid system that can be configured to mimic real microgrids as a scaled-down version of a real system. It provides a platform that enables to extrapolate how the real microgrid under consideration will perform, thus de-risking the installation and commissioning of the microgrid and providing greater confidence in the commissioning process. The facility can also be used for testing and evaluation of different battery, electrolyser and fuel cell technologies and their integration into grid systems.

The following services are offered through REMHART:

1. Performance evaluation of batteries, electrolysers, and fuel cell technologies.
2. Modelling and analysis of hybrid power systems
3. Development and testing of control strategies and algorithms for hybrid power systems
4. Integration of solar PV, hydrogen electrolyser and fuel cells into the grid
5. Techno-economic analysis of renewable power systems
6. Identifying, analysing, evaluating and addressing the cyber security threats related to microgrids.
7. Penetration Testing Service - to identify and verify the vulnerabilities affecting microgrid systems.
8. IoT sensors for remote monitoring and predictive maintenance of microgrids.

Contact us

Suresh Thennadil

Director, Energy and Resources Institute
College of Engineering, IT & Environment

T: +61 8 8946 6564

E: suresh.thennadil@cdu.edu.au